

MANTUO ORIGINAL TEACHING NOTES / 2026.07-v2

TMUA Foundations Review Notes

TMUA 基础复习笔记

From exam structure and curriculum gaps to core mathematics, timed reasoning and paper review

从考试结构、课程缺口与核心数学，到限时推理和真题复盘

The logo consists of the letters 'MT' in a white, bold, sans-serif font, centered within a solid purple square.

MANTUO ADMISSION TEST LIBRARY

不再为升学考试而焦虑

Original teaching notes · Not an official UAT-UK publication

How to Use This Pack

如何使用这份笔记

This release covers exam orientation, non-calculator strategy, logic and proof, algebra and functions, sequences and geometry, trigonometry, exponentials and logarithms, calculus, mathematical reasoning, paper review, and A-Level, IB and AP bridges.

本版包含考试定位、无计算器策略、逻辑与证明、代数与函数、数列与几何、三角、指数对数、微积分、数学推理、真题复盘，以及 A-Level、IB、AP 课程衔接。

EDITION	Foundation Edition: structured for first-pass review, curriculum mapping and active recall. 基础版：用于第一轮复习、课程映射与主动回忆。
AUTHORSHIP BOUNDARY	This resource independently reorganises and explains the public exam framework. All examples, checkpoint questions and explanations are original Mantou work and are not official questions; official materials are used only to verify scope and facts. 本资料依据公开考试框架重新组织并原创撰写。例题、自测题与解析均非官方真题；官方资料只用于核对考试范围与事实。
GO DEEPER	After the Foundation Edition, use the Six-Week Precision Training Plan to structure the work, then collect accuracy, timing and error evidence in past papers. Worked Explanations identify the reasoning worth repairing first. 完成基础版后，使用六周精确训练计划安排节奏，并在历年真题中积累正确率、用时和错误类型证据；逐题深度解析用于定位最值得修复的思路。

FACTS FIRST

Official Exam Map

官方考试地图

TWO PAPERS 两张试卷 75 minutes each Paper 1 与 Paper 2 各 75 分钟	QUESTION COUNT 题量 20 multiple-choice questions per paper 每张 20 道选择题，共 40 道
SCORING 计分 Equal weighting; no penalty for incorrect answers 所有题等权；答错不倒扣；应尝试全部题目	TOOLS 工具 No calculator and no formula booklet 不可使用计算器；不提供公式表
APPLICATIONS Paper 1 Applications of Mathematical Knowledge 应用 Section 1 数学知识解决不同情境的问题	REASONING Paper 2 Mathematical Reasoning 同时使用 Section 1 数学与 Section 2 逻辑、证明和辨错

MANTUO THREE-PASS METHOD

Pass 1: Secure	Pass 2: Resolve	Final check: Complete
第一轮：确定能做	第二轮：限制分支	最后检查：完成全部
Take the questions with a clear route first; flag uncertain starts.	Write constraints, exact values or a counterexample target before calculating.	Answer every question, then check signs, intervals, wording and transcription.
先拿下路径清晰的题。若读完仍没有可执行的第一步，标记后继续。	对已标记题写出约束、精确值或反例目标，减少无方向试算。	考试不倒扣。确认每题都已作答，再检查符号、范围、题目问法和误读。

A-LEVEL / IB / AP

Curriculum Bridge

课程衔接：具体缺什么

Curriculum mapping estimates syllabus exposure; it does not replace an individual diagnosis. Completed units, examination board and timed evidence can change the conclusion.

课程映射只判断 syllabus exposure，不代替个人能力诊断。完成单元、考试局和真实做题证据会改变结论。

A-Level / IAL Mathematics	STRONG START 通常覆盖较多，但不是自动全部覆盖
LIKELY COVERED	- Algebra, functions and graphs 代数、函数与图像 - Trigonometry, exponentials and logarithms 三角、指数与对数 - Foundation calculus 基础微积分 - Sequences and coordinate geometry 数列与坐标几何
CHECK GAPS	- Whether the corresponding Pure Mathematics units are complete 是否已经完成对应 Pure Mathematics 单元 - Paper 2 necessary and sufficient conditions, negating quantifiers, contradiction and counterexamples Paper 2 的 necessary / sufficient、量词否定、反证法与反例 - GCSE-level circle theorems, units, ratios and exact calculation GCSE 层面的圆定理、单位、比例与精确计算
FIRST ACTION	Map the curriculum first. If the core Pure units are complete, prioritise logical language, exact non-calculator work and timed multiple-choice practice. 先做课程映射；若核心 Pure 单元已完成，把时间优先用于逻辑语言、无计算器精确运算和限时选择题。

IB Mathematics: Analysis and Approaches HL	STRONG START 数学主干通常较强，需补英国考试语言与非计算器节奏
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LIKELY COVERED	- Functions, trigonometry and sequences 函数、三角、数列 - Calculus and graph analysis 微积分与图像分析 - Algebraic modelling 代数建模
CHECK GAPS	- Circle theorems and UK GCSE supporting knowledge 圆定理与英国 GCSE supporting knowledge - Formal logic vocabulary and evaluation of proofs formal logic vocabulary 与证明辨错 - Exact values, factorisation and option discrimination without a GDC 不用 GDC 的精确值、因式分解和选项辨析
FIRST ACTION	Do not relearn the entire mathematics course. Audit geometry and logic gaps, then build speed through TMUA question formats. 不要重复学习整门数学；先核对几何与逻辑缺口，再用 TMUA 题型训练速度。

IB Mathematics: Analysis and Approaches SL	PARTIAL 部分覆盖；完成单元决定实际缺口
LIKELY COVERED	- Foundation functions and graphs 基础函数与图像 - Foundation trigonometry and calculus 基础三角和微积分 - Foundation statistics and algebra 基础统计与代数
CHECK GAPS	- Deeper algebraic manipulation, exponentials, logarithms and trigonometric equations 更深的代数变形、指数对数与三角方程 - Sequences, coordinate geometry and combined graph work 数列、坐标几何和图像综合 - All Paper 2 logic and proof Paper 2 全部逻辑与证明
FIRST ACTION	Build a topic-by-topic learning list first; do not consume a complete past paper merely to locate all gaps. 先按知识点做补学清单，不建议直接用整套真题定位所有缺口。

IB Mathematics: Applications and Interpretation HL	PARTIAL 应用与技术工具经验有帮助，但纯数学与无计算器要求需单独核对
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LIKELY COVERED	- Modelling and data interpretation 建模与数据解释 - Some functions and calculus 部分函数与微积分 - Graph understanding with technology 技术工具下的图像理解
CHECK GAPS	- Exact algebra, factorisation and surds 精确代数、因式分解与根式 - Trigonometric identities, sequences and coordinate geometry 三角恒等式、数列和坐标几何 - Non-calculator graph reasoning, formal logic and proof 无计算器图像推理、形式逻辑与证明
FIRST ACTION	Take a topic diagnostic without technology, repair exact algebra first, and then move into Paper 2. 把技术工具拿掉后做一次分主题诊断；先补精确代数，再进入 Paper 2。

AP Precalculus + AP Calculus AB/BC	PARTIAL 函数与微积分可衔接，但 AP 课程组合不能直接等同 TMUA 范围
LIKELY COVERED	- Functions, graphs and trigonometry 函数、图像与三角 - Exponentials and logarithms 指数对数 - Differentiation and integration, depending on AB or BC progress 微分与积分（取决于 AB / BC 进度）
CHECK GAPS	- Surds, binomial work, finite geometric series and UK-style exact answers 根式、二项式、有限几何级数与英国式精确答案 - Circle theorems, coordinate geometry and GCSE supporting knowledge 圆定理、坐标几何与 GCSE supporting knowledge - Necessary and sufficient conditions, quantifiers, counterexamples and proof evaluation necessary / sufficient、量词、反例和证明辨错
FIRST ACTION	Audit Algebra II and Geometry records as well; AP Calculus alone does not demonstrate that prerequisite algebra and geometry are covered. 同时核对 Algebra II / Geometry 学习记录；仅有 AP Calculus 不能证明前置代数与几何已经覆盖。

Build the Exam Map First

先建立考试地图

TMUA is not simply a contest in how much mathematics you know. It tests whether you can remain precise without a calculator, under fixed timing and among closely matched options.

TMUA 不是单纯比谁会更多数学，而是看你能否在无计算器、固定时间和相似选项中保持精确。

LEARNING OUTCOMES

- Distinguish Paper 1 applications from Paper 2 mathematical reasoning

能区分 Paper 1 的知识应用与 Paper 2 的数学推理

- Classify a failed start as a knowledge, route, execution or reading error

能把做不出题的原因分成知识缺口、路径缺口、运算错误和读题错误

- Use a three-pass strategy to complete the paper rather than exhausting time on the first hard question

能用三轮策略完成全部题，而不是在第一道难题上耗尽时间

01.1

Diagnose the Bottleneck

先判断卡在哪里

The same wrong answer can require completely different repairs. Not knowing the discriminant is a knowledge gap; knowing it but failing to activate it is a route gap; correct reasoning with a sign error is an execution error; reversing only if is a reading error.

同一道错题，补救方式可能完全不同。不会判别式属于 knowledge gap；知道判别式但想不到使用属于 route gap；推导正确但符号出错属于 execution error；把 only if 看反属于 reading error。

Recording only 'wrong' has no training value. At minimum, record the topic, the first error, active-thinking time, whether the answer changed, and the next action.

每次复盘只记录“错了”没有训练价值。至少记录：知识点、第一处错误、主动思考时间、是否改答、下一次动作。

Knowledge gap 知识缺口	The concept, formula or condition of use is unknown. Learn one minimum knowledge unit next. 概念、公式或适用条件本身不知道。下一步是学习一个最小知识单元。
Route gap 路径缺口	The knowledge is available but the conditions do not trigger a start. Record the signal and first move next. 知识会，但无法从题目条件启动。下一步是总结触发信号和第一步。
Execution error 执行错误	The route is correct but algebra, sign, range or transcription fails. Add a checkpoint next. 思路正确但代数、符号、范围或抄写错误。下一步是建立检查点。
Reading error 读题错误	Necessary versus sufficient, at least versus at most, integer versus real, or the requested quantity is misread. Circle the qualifier next. 误读必要/充分、至少/至多、整数/实数或题目所求。下一步是圈定限定词。

ACTIVE RECALL / 合上笔记回答

You can redo the question but could not find a starting move. Which error type is this?

一道题你会完整重做，但第一次完全想不到从哪里开始，它属于哪类错误？

ANSWER / A route gap. Record the trigger and the first move.

答案 / 路径缺口（route gap）。记录触发信号和第一步，而不是重新抄完整解答。

01.2

Non-calculator Does Not Mean Mental Arithmetic Only

无计算器不等于只练心算

The real skills are exact form, structured cancellation, estimation bounds and option validation. Expanding every expression is often slower than preserving factors or symmetry.

真正需要训练的是 exact form、结构化约分、估算边界和选项验证。把所有表达式都展开，往往比保留因式或对称结构更慢。

With five options, ask whether the question requires a complete solution or only the rejection of four impossibilities. The options are part of the information.

遇到五个选项时，先问：题目要求完整求解，还是只需要排除四个不可能？选项是题目信息的一部分。

Exact form 精确形式	Preserve fractions, surds and π ; use safe estimation only when comparing magnitude. 优先保留分数、根式与 π ；仅在比较大小时使用安全估算。
Structure first 先看结构	Look for common factors, conjugates, symmetry, the discriminant and graph meaning before deciding to expand. 先找公因式、共轭、对称、判别式和图像意义，再决定是否展开。
Option-aware 利用选项	Substitution, parity, sign, order of magnitude and bounds can eliminate options quickly. 代入、奇偶性、符号、数量级与边界都可以快速排除选项。

ACTIVE RECALL / 合上笔记回答

Why should 3 minutes 45 seconds not become a rigid per-question limit?

为什么“每题平均 3 分 45 秒”不能变成每题强制计时？

ANSWER / Question difficulty varies; the average guides the paper, not every item.

答案 / 题目难度和路径长度不同。平均值用于整卷节奏，不代表每题必须平均分配。

Logic, Proof and Counterexamples

逻辑、证明与反例

Paper 2 is not about memorising symbols. Translate English statements into reliable conditional directions and know how to disprove a universal claim.

Paper 2 的关键不是背符号，而是把英文句子稳定翻译成条件方向，并知道怎样推翻一个全称结论。

LEARNING OUTCOMES

- Distinguish an implication, its converse and its contrapositive

能区分 implication、converse 与 contrapositive

- Translate necessary and sufficient into the correct direction

能把 necessary 和 sufficient 转换成正确方向

- Negate for all and there exists, then choose direct proof, cases, contradiction or counterexample

能否定 for all / there exists，并选择直接证明、分类、反证或反例

02.1

Condition Direction: Draw the Arrow

条件方向：先画箭头

Let A be the condition and B the conclusion. A is sufficient for B means A guarantees B, so $A \Rightarrow B$. A is necessary for B means B cannot hold without A, so $B \Rightarrow A$.

把 A 当作条件，B 当作结论。A is sufficient for B 表示 A 足以保证 B，因此方向是 $A \Rightarrow B$ 。A is necessary for B 表示 B 成立必须有 A，因此方向是 $B \Rightarrow A$ 。

The implication $A \Rightarrow B$ is equivalent to its contrapositive $\neg B \Rightarrow \neg A$. Its converse $B \Rightarrow A$ need not be true.

原命题 $A \Rightarrow B$ 与逆否命题 $\neg B \Rightarrow \neg A$ 等价；逆命题 $B \Rightarrow A$ 不一定成立。

Sufficient 充分	A guarantees B. A is sufficient for B $A \Rightarrow B$
Necessary 必要	B requires A. A is necessary for B $B \Rightarrow A$
Contrapositive 逆否	Logically equivalent to the original implication. 与原命题逻辑等价 $A \Rightarrow B \Leftrightarrow \neg B \Rightarrow \neg A$
Converse 逆命题	Reverses the direction and cannot be assumed true. 把方向反过来，不能自动认为正确 $B \Rightarrow A$

PROBLEM	Let n be an integer. Prove that if 4 divides n^2, then n is even. 设 n 为整数。证明：如果 n^2 能被 4 整除，那么 n 是偶数。
Write the contrapositive 1. 写逆否	Prove that if n is not even, then n^2 is not divisible by 4. For an integer, not even means odd. 证明“如果 n 不是偶数，那么 n^2 不能被 4 整除”。对整数而言，不是偶数就是奇数。
Represent an odd integer 2. 表示奇数	Let $n = 2k + 1$ for some integer k. 令 $n = 2k + 1$，其中 k 为整数。 $n = 2k + 1$
Square and inspect the remainder 3. 平方并看余数	$n^2 = 4k^2 + 4k + 1 = 4k(k + 1) + 1$, so n^2 leaves remainder 1 on division by 4. $n^2 = 4k^2 + 4k + 1 = 4k(k+1)+1$，所以 n^2 除以 4 余 1。 $n^2 = 4k(k + 1) + 1$
Return to the implication 4. 回到原命题	The contrapositive is true, so the original implication is true. 逆否命题成立，因此原命题成立。
CONCLUSION	The key is not to take a square root of $n^2 = 4m$, but to use parity to obtain a rigorous remainder statement. 关键不是从 $n^2 = 4m$ 直接“开方”，而是使用奇偶分类得到严格的余数结论。
COMMON TRAP	Writing that divisibility of n^2 by 4 directly implies divisibility of n by 2 assumes the result that must be proved. 从 n^2 可被 4 整除直接写 n 可被 2 整除，等于把要证明的结论当作已知。

ACTIVE RECALL / 合上笔记回答

Over the reals, is $x > 3$ sufficient or necessary for $x^2 > 9$?

“ $x > 3$ 是 $x^2 > 9$ 的充分条件”在实数范围内正确吗？必要吗？

ANSWER / Sufficient but not necessary, because $x < -3$ also gives $x^2 > 9$.

答案 / 充分但不必要。 $x > 3$ 一定推出 $x^2 > 9$ ，但 $x < -3$ 也会使 $x^2 > 9$ 。

02.2

Quantifiers and Counterexamples

量词与反例

To negate 'every object satisfies P ', it is enough that at least one object does not satisfy P . To negate 'there exists an object satisfying P ', every object must fail P .

否定“所有对象都满足 P ”时，只需存在一个对象不满足 P 。否定“至少存在一个对象满足 P ”时，需要所有对象都不满足 P 。

One valid counterexample disproves a universal statement; many supporting examples still do not prove it.

要推翻全称命题，一个合法反例就足够；列出很多支持例子仍然不能证明全称命题。

Not ‘for all’ 否定 “对所有”	Becomes ‘there exists at least one object that does not satisfy it’. 变成 “至少存在一个不满足” $\text{not}(\text{for every } x, P) \Leftrightarrow \text{some } x \text{ is not } P$
Not ‘there exists’ 否定 “存在”	Becomes ‘every object fails to satisfy it’. 变成 “所有都不满足” $\text{not}(\text{some } x \text{ is } P) \Leftrightarrow \text{every } x \text{ is not } P$
Counterexample 反例	It must satisfy the premise while making the conclusion fail. 必须满足命题的前提，同时让结论失败。
Illegal division 非法除法	Before dividing by an expression containing a variable, establish that it is non-zero. 除以含变量表达式之前，必须确认它不为 0。

ACTIVE RECALL / 合上笔记回答

What form must the negation of ‘every even number is the sum of two primes’ take?

否定 “所有偶数都能写成两个质数之和” 时，你需要证明什么形式的句子？

ANSWER / There exists an even number that cannot be written as the sum of two primes.

答案 / 存在至少一个偶数，不能写成两个质数之和。这里只是在转换逻辑形式，不是在断言该反例实际存在。

CHAPTER 03 / TMUA FOUNDATIONS

Algebra, Functions and Graphs

代数、函数与图像

This chapter builds four reusable TMUA actions rather than complicated tricks: state restrictions, preserve structure, connect algebra to graphs, and verify roots.

这一章不追求复杂技巧，而是建立 TMUA 最常复用的四个动作：先写限制、保留结构、连接图像、最后验根。

LEARNING OUTCOMES

- State valid conditions before squaring, taking roots, cancelling or applying logarithms
能在平方、开方、约分和取对数前写出合法条件
- Use the discriminant to determine the number of roots and parameter range
能用判别式判断根的数量与参数范围
- Recognise graph transformations and interpret intersections as equation solutions
能识别函数图像变换并把交点解释为方程的解

03.1

State Restrictions Before Manipulating

先写定义域与等价条件

Many errors come from treating a one-way implication as an equivalent transformation. Squaring can introduce extraneous roots, cancellation can lose a zero-denominator restriction, and radicals or logarithms require a domain check first.

很多错误不是算错，而是把单向推导当成等价变形。平方可能引入增根；约分可能丢掉使分母为零的限制；根式和对数必须先检查定义域。

Whenever an operation can change the solution set, write a check beside it: domain, denominator, sign or interval.

每次做可能改变解集的动作，在草稿边写一个 check：domain、denominator、sign 或 interval。

Square root 平方根	$\sqrt{g(x)}$ requires $g(x) \geq 0$; if $\sqrt{g(x)} = h(x)$, then $h(x)$ must also be non-negative. $\sqrt{g(x)}$ 要求 $g(x) \geq 0$ ；若 $\sqrt{g(x)} = h(x)$ ，还必须有 $h(x) \geq 0$ 。
Denominator 分母	Record that the cancelled expression is non-zero before cancellation. 约分前记录被除表达式不为 0。
Logarithm 对数	log base a of $g(x)$ requires $a > 0$, $a \neq 1$ and $g(x) > 0$. $\log_a g(x)$ 要求 $a > 0$ 、 $a \neq 1$ 且 $g(x) > 0$ 。
Verification 验根	Candidate roots from equations involving radicals, fractions or squaring must be checked in the original equation. 含根式、分式或平方步骤的方程，候选解必须代回原式。

ORIGINAL WORKED EXAMPLE	
Original Worked Example: Restrict Before Squaring	
原创例题：根式方程先限制再平方	
PROBLEM	Solve $\sqrt{x+3} = x-1$ over the reals. 在实数范围内解 $\sqrt{x+3} = x-1$ 。
State the restriction 1. 写限制	The left side is non-negative, so $x-1 \geq 0$ and $x \geq 1$. This also guarantees $x+3 \geq 0$. 左侧非负，因此 $x-1 \geq 0$ ，即 $x \geq 1$ 。这个条件同时保证 $x+3 \geq 0$ 。 $x \geq 1$
Square both sides 2. 平方	$x+3 = (x-1)^2$, which rearranges to $x^2 - 3x - 2 = 0$. $x+3 = (x-1)^2$ ，整理得 $x^2 - 3x - 2 = 0$ 。 $x^2 - 3x - 2 = 0$
Find candidate roots 3. 求候选解	The quadratic formula gives $x = (3 \pm \sqrt{17}) / 2$. 使用求根公式得到 $x = (3 \pm \sqrt{17}) / 2$ 。 $x = (3 \pm \sqrt{17}) / 2$
Filter using the restriction 4. 用限制筛选	$(3 - \sqrt{17}) / 2$ is negative and violates $x \geq 1$, so only the positive root remains. $(3 - \sqrt{17}) / 2 < 0$ ，不满足 $x \geq 1$ ；只保留正根。
CONCLUSION	The unique real solution is $x = (3 + \sqrt{17}) / 2$. 唯一实数解为 $x = (3 + \sqrt{17}) / 2$ 。

COMMON TRAP	Both roots after squaring are only candidates; failing to check the original equation retains an extraneous root. 平方后的两个根只是候选解；不检查原式会保留增根。
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ACTIVE RECALL / 合上笔记回答

Before solving $1/(x-2) = 3/(x^2-4)$, which values must be excluded?

解 $1/(x-2) = 3/(x^2-4)$ 时，开始计算前必须记录哪些排除值？

ANSWER / $x \neq 2$ and $x \neq -2$.

答案 / $x \neq 2$ 且 $x \neq -2$ ，因为原方程的两个分母都不能为 0。

03.2

Discriminant and Graph: Two Languages for One Structure

判别式与图像是同一件事的两种语言

For $ax^2 + bx + c = 0$, the discriminant $\Delta = b^2 - 4ac$ gives the number of x-axis intersections. Parameter questions often require only translating two distinct real roots, tangency or no real roots into a condition on Δ .

对二次方程 $ax^2 + bx + c = 0$ ，判别式 $\Delta = b^2 - 4ac$ 告诉你图像与 x 轴交点的数量。参数题常常无需先求根，只需把“两个不同实根”“相切”或“无实根”翻译成 Δ 的条件。

Intersections of $f(x)$ and $g(x)$ correspond to solutions of $f(x) = g(x)$. Estimate the possible number of roots graphically before algebra to catch missing solutions.

两个函数 $f(x)$ 与 $g(x)$ 的交点，对应方程 $f(x) = g(x)$ 的解。先看图像可能根数，再做代数，可以及时发现漏解。

Two distinct real roots 两个不同实根	The discriminant is strictly positive. 判别式严格大于 0 $\Delta = b^2 - 4ac > 0$
Repeated real root 重根 / 相切	The discriminant is zero. 判别式等于 0 $\Delta = 0$
No real roots 无实根	The discriminant is negative. 判别式小于 0 $\Delta < 0$

ORIGINAL WORKED EXAMPLE

Original Worked Example: Parameter and Distinct Roots

原创例题：参数与不同实根

PROBLEM

The equation $x^2 - 2(k+1)x + k^2 + 1 = 0$ has two distinct real roots. Find the range of k .
方程 $x^2 - 2(k+1)x + k^2 + 1 = 0$ 有两个不同实根。求 k 的范围。

Translate the condition 1. 翻译题意	Two distinct real roots means $\Delta > 0$. 两个不同实根意味着 $\Delta > 0$ 。
Calculate while preserving structure 2. 计算但保留结构	$\Delta = [-2(k+1)]^2 - 4(k^2 + 1) = 8k$. $\Delta = [-2(k+1)]^2 - 4(k^2+1) = 8k$. $\Delta = 4(k+1)^2 - 4(k^2 + 1) = 8k$
Solve the inequality 3. 解不等式	$8k > 0$, so $k > 0$. $8k > 0$, 因此 $k > 0$. $k > 0$
CONCLUSION	$k > 0$. There is no need to calculate the two roots. $k > 0$. 无需先写出两个根。
COMMON TRAP	Two real roots is not the same as two distinct real roots; the latter requires a strict inequality. “两个实根”与“两个不同实根”不同；后者要求严格不等式。

ACTIVE RECALL / 合上笔记回答

If $y=f(x)$ and $y=2x+1$ meet three times, how many distinct real roots does $f(x)=2x+1$ have?

若 $y=f(x)$ 与 $y=2x+1$ 有三个交点，方程 $f(x)=2x+1$ 有多少个不同实根？

ANSWER / Three, one x-coordinate per intersection.

答案 / 三个。每一个交点的 x 坐标对应一个不同实根。

03.3

Graph Transformations: Inside Changes Run Opposite

图像变换：括号内方向相反

An external change acts directly on output y . A change inside the brackets acts on input x , so the direction appears reversed. Verify with a known point rather than memorising a slogan.

外部变化直接作用在输出 y 上；括号内部变化作用在输入 x 上，因此平移方向看起来相反。不要只背口诀：用一个已知点代入新表达式验证位置。

For example, $(a, f(a))$ on f becomes $(a + 3, f(a))$ on $y = f(x - 3)$, so the graph shifts three units right.

例如 $f(x)$ 上点 $(a, f(a))$ 在 $y = f(x-3)$ 上会变成 $(a+3, f(a))$ ，所以整体向右 3。

Vertical shift 竖直平移	$y = f(x) + a$ shifts up by a when $a > 0$. $y = f(x) + a$: 向上 a ($a > 0$)
Horizontal shift 水平平移	$y = f(x + a)$ shifts left by a when $a > 0$. $y = f(x + a)$: 向左 a ($a > 0$)
Vertical scale 竖直伸缩	$y = a f(x)$ multiplies every y -coordinate by a . $y = a f(x)$: 所有 y 坐标乘 a

Horizontal scale | 水平伸缩

$y = f(ax)$ divides every x -coordinate by a .

$y = f(ax)$: 所有 x 坐标除以 a

ACTIVE RECALL / 合上笔记回答

How can the horizontal change from $y=f(x)$ to $y=f(2x-3)$ be described?

把 $y = f(x)$ 变成 $y = f(2x-3)$, 可怎样描述水平变化?

ANSWER / Write $f(2(x-3/2))$: horizontal scale factor $1/2$, with a shift right by $3/2$.

答案 / 写成 $f(2(x-3/2))$: 先理解为水平坐标缩为原来的 $1/2$, 再向右平移 $3/2$ 。用特征点验证, 不要只凭顺口诀。

CHAPTER 04 / TMUA FOUNDATIONS

Sequences, Coordinate Geometry and Trigonometry

数列、坐标几何与三角

These questions compress several familiar ideas into one multiple-choice problem. Identify the structure before choosing a term formula, distance, gradient or exact trigonometric value.

这些题经常把多个熟悉知识点压缩到同一道选择题里。关键是先辨认结构, 再选择通项、距离、斜率或精确三角值。

LEARNING OUTCOMES

- Choose an appropriate representation for arithmetic, geometric and recursive sequences

能在等差、等比与递推数列之间选择合适表示

- Translate lines, circles and tangency into distance or discriminant conditions

能把直线、圆与相切条件转换成距离或判别式条件

- Use exact trigonometric values, identities and quadrant signs to eliminate impossible options

能使用精确三角值、恒等式和象限符号排除不可能选项

04.1

Sequences: Identify the Model Before Using a Formula

数列: 先分清模型, 再代公式

Arithmetic sequences have a fixed difference; geometric sequences have a fixed ratio. When non-adjacent terms are given, do not assume the ratio is positive. From r^2 , retain both signs until monotonicity, sign or range information selects one.

等差数列关注固定差, 等比数列关注固定倍数。题目给出不相邻项时, 不要默认公比为正; 由 r^2 得到 r 时必须结合题目中的正负、单调或范围条件筛选。

A recurrence describes how neighbouring terms change; an explicit formula gives the n th term. Writing early terms can reveal structure, but the observed pattern still needs recurrence or algebraic verification.

递推式告诉你相邻项怎样变化, 通项告诉你第 n 项是什么。选择题中可先写出前几项找结构, 但观察到的规律仍要用递推或代数验证。

Arithmetic 等差	The nth term equals the first term plus $n - 1$ common differences. 第 n 项等于首项加 $(n-1)$ 个公差。 $u_n = a + (n - 1)d$
Geometric 等比	The nth term equals the first term multiplied by the common ratio to the power $n - 1$. 第 n 项等于首项乘公比的 $n-1$ 次方。 $u_n = ar^{(n - 1)}$
Finite geometric sum 有限等比和	When $r \neq 1$, preserve the factored form where possible. r 不等于 1 时优先保留因式结构。 $S_n = a(1 - r^n)/(1 - r)$
Sign check 符号检查	From $r^2 = c$, write $r = \pm\sqrt{c}$; only additional conditions can remove a branch. 从 $r^2 = c$ 得到 $r = \pm\sqrt{c}$; 只有额外条件才能删掉其中一个分支。

ORIGINAL WORKED EXAMPLE Original Worked Example: Use the Sign Condition 原创例题：不要漏掉公比的符号条件	
PROBLEM	A geometric sequence has positive terms, first term 3 and third term 12. Find the sum of its first four terms. 一个各项均为正数的等比数列首项为 3，第三项为 12。求前四项之和。
Use the third term 1. 用第三项	The third term is $3r^2$, so $3r^2 = 12$ and $r^2 = 4$. 第三项为 $3r^2$，因此 $3r^2 = 12$，得到 $r^2 = 4$。 $3r^2 = 12$
Apply the positivity condition 2. 使用正数条件	Algebra gives $r = \pm 2$, but all terms are positive, so r cannot be -2 and must equal 2. 代数上 $r = \pm 2$；但所有项均为正，所以公比不能是 -2，必须取 $r = 2$。 $r = 2$
Sum the terms 3. 求和	The first four terms are 3, 6, 12 and 24, with total 45. 前四项为 3、6、12、24，因此总和为 45。 $S_4 = 45$
CONCLUSION	The sum of the first four terms is 45. 前四项之和为 45。
COMMON TRAP	Writing only $r = \sqrt{4}$ hides an algebraic branch. Write ± 2 first, then use the condition to select. 只写 $r = \sqrt{4}$ 会掩盖一个代数分支；正确做法是先写 ± 2，再用题目条件筛选。

If a geometric sequence has first term a and fourth term $-8a$, with $a \neq 0$, what is the common ratio?

若等比数列首项为 a ，第四项为 $-8a$ ，且 $a \neq 0$ ，公比是多少？

ANSWER / $r^3 = -8$, so $r = -2$.

答案 / $r^3 = -8$ ，所以 $r = -2$ 。三次方在实数范围内只有一个对应根。

04.2

Lines, Circles and Trigonometry: Translate the Diagram

直线、圆与三角：把图形翻译成条件

A coordinate diagram carries conditions. Parallel means equal gradients, perpendicular means gradient product -1 where defined, and tangent means centre-to-line distance equals radius or the simultaneous equations have discriminant zero.

坐标几何题的图形只是信息载体。平行转成斜率相同，垂直转成斜率乘积为 -1 （斜率存在时），相切转成圆心到直线距离等于半径，或联立后判别式等于 0 。

In trigonometry, prefer exact values and quadrant signs. When $\sin^2\theta + \cos^2\theta$, $\tan\theta = \sin\theta/\cos\theta$ or supplementary-angle relations appear, simplify the structure before finding an angle.

三角题优先使用精确值和象限符号。看到 $\sin^2\theta + \cos^2\theta$ 、 $\tan\theta = \sin\theta/\cos\theta$ 或补角关系时，先化结构，再决定是否求角。

Circle 圆	Standard form for centre (a, b) and radius r. 圆心 (a,b)、半径 r 的标准式。 $(x - a)^2 + (y - b)^2 = r^2$
Perpendicular 垂直	For two non-vertical perpendicular lines, the gradient product is -1. 两条非竖直线垂直时，斜率乘积为 -1。 $m_1 m_2 = -1$
Tangent 相切	The perpendicular distance from the centre to the line equals the radius; equivalently, the simultaneous equations have one intersection. 圆心到直线的垂直距离等于半径；等价地，联立方程只有一个交点。
Exact trigonometry 精确三角	Keep surds and fractions for 0°, 30°, 45°, 60°, 90° and related angles rather than converting to decimals. 对 0°、30°、45°、60°、90° 度及其相关角保留根式和分数，不先转小数。

ORIGINAL WORKED EXAMPLE

Original Worked Example: Identify Tangents by Distance

原创例题：用距离识别切线

PROBLEM

The line $y = mx + 5$ is tangent to the circle $x^2 + y^2 = 5$. Find all possible values of m .

直线 $y = mx + 5$ 与圆 $x^2 + y^2 = 5$ 相切。求 m 的所有可能值。

Identify centre and radius 1. 写圆心和半径	The centre is the origin and the radius is $\sqrt{5}$. 圆心是原点，半径为 $\sqrt{5}$ 。
Write the line in standard form 2. 写直线标准式	The line is $mx - y + 5 = 0$. Its distance from the origin is $5/\sqrt{m^2 + 1}$. 直线可写成 $mx - y + 5 = 0$ 。原点到该直线的距离是 $5/\sqrt{m^2+1}$ 。 $d = 5/\sqrt{m^2 + 1}$
Apply the tangency condition 3. 使用相切条件	Equate the distance to the radius and square: $25/(m^2 + 1) = 5$, so $m^2 = 4$. 令距离等于半径并平方，得到 $25/(m^2+1) = 5$ ，所以 $m^2 = 4$ 。 $25/(m^2 + 1) = 5$
Keep both directions 4. 保留两个方向	Hence $m = 2$ or $m = -2$; both symmetric lines are tangent to the circle. 因此 $m = 2$ 或 $m = -2$ ；两条对称直线都与圆相切。
CONCLUSION	$m = 2$ or $m = -2$. $m = 2$ 或 $m = -2$ 。
COMMON TRAP	Tangency does not force a positive gradient. Symmetry about the y-axis suggests two symmetric answers. 相切不代表斜率只有正值；圆关于 y 轴对称，因此应预期两个对称答案。

ACTIVE RECALL / 合上笔记回答

If $\cos \theta$ is negative and $\tan \theta$ is positive, which quadrant contains θ between 0 and 360 degrees?

若 $\cos \theta < 0$ 且 $\tan \theta > 0$ ， θ 在 0° 到 360° 的哪个象限？

ANSWER / Quadrant III: sine and cosine are both negative.

答案 / 第三象限。tan 为正表示 sin 与 cos 同号；cos 为负，所以二者都为负。

Exponentials, Logarithms and Calculus Models

指数、对数与微积分模型

TMUA tests not only differentiation and integration, but also domains, growth behaviour, and the meaning of derivatives and areas in context.

TMUA 不只考能否求导积分，更考你是否理解函数的定义域、增长方式，以及导数和面积在题目情境中的含义。

LEARNING OUTCOMES

- Use a common base, monotonicity and domain to solve exponential and logarithmic equations

能用同底、单调性与定义域处理指数和对数方程

- Interpret derivatives as gradient, increase or decrease, and extremum conditions

能把导数解释为斜率、增减和极值条件

- Interpret a definite integral as signed area and check the valid range of a model

能把定积分解释为带符号面积，并检查模型的有效范围

05.1

Exponentials and Logarithms: Use One Language

指数与对数：先统一语言

If both sides can be written as positive powers of the same base, one-to-one behaviour is usually faster than taking logarithms. If logarithms are necessary, check positive arguments and never split $\log(a + b)$ into $\log a + \log b$.

若两边能改写成同底正数幂，利用指数函数的一一对应性通常比直接取对数更快。若必须取对数，先确认真数为正，并避免把 $\log(a+b)$ 错拆成 $\log a + \log b$ 。

For growth and decay, write the model before solving and then check the variable range. A correct algebraic solution may still represent negative time or a probability outside 0 to 1.

增长与衰减问题先写模型，再检查变量范围。代数解即使正确，也可能落在时间为负或概率超出 0 到 1 的无效区间。

Same base 同底	For $a > 0$ and $a \neq 1$, $a^u = a^v$ implies $u = v$. 当 $a > 0$ 且 $a \neq 1$ 时，$a^u = a^v$ 推出 $u = v$。 $a^u = a^v \Rightarrow u = v$
Log domain 对数定义域	$\log g(x)$ has a real value only when $g(x) > 0$. $\log g(x)$ 只在 $g(x) > 0$ 时有实数意义。
Product law 乘积法则	A product can split into a sum of logarithms; a sum cannot. 只有乘积可以拆成和；和不能这样拆。 $\log(ab) = \log a + \log b$
Model range 模型范围	Return the solution to the practical range of the contextual variable, not only the algebraic expression. 最后把解代回题目变量的实际范围，而不仅是原代数式。

原创例题：先改写同底

PROBLEM	Solve $2^{(x+1)} = 8 \times 4^x$. 解方程 $2^{(x+1)} = 8 \times 4^x$。
Rewrite everything as powers of 2 1. 全部改写成 2 的幂	$8 = 2^3$ and $4^x = 2^{(2x)}$, so the right side is $2^{(2x+3)}$. $8 = 2^3$, $4^x = 2^{(2x)}$, 所以右侧是 $2^{(2x+3)}$。 $8 \times 4^x = 2^{(2x+3)}$
Equate the exponents 2. 比较指数	The base 2 is positive and not 1, so $x + 1 = 2x + 3$. 底数 2 为正且不等于 1, 因此 $x + 1 = 2x + 3$。 $x + 1 = 2x + 3$
Solve and verify 3. 求解并检查	This gives $x = -2$; substitution makes both sides equal to $1/2$. 得到 $x = -2$; 代入两边都等于 $1/2$。
CONCLUSION	$x = -2$. $x = -2$。
COMMON TRAP	$4^x = (2^2)^x = 2^{(2x)}$, not $2^{(x+2)}$. $4^x = (2^2)^x = 2^{(2x)}$, 不是 $2^{(x+2)}$。

ACTIVE RECALL / 合上笔记回答

Why must $x=2$ still be checked after solving $\log(x-1)=\log(3-x)$?为什么 $\log(x-1) = \log(3-x)$ 求出 $x = 2$ 后仍需检查?ANSWER / Both logarithm arguments must be positive; $x=2$ lies in $1 < x < 3$.答案 / 原式要求 $x-1 > 0$ 且 $3-x > 0$, 即 $1 < x < 3$ 。 $x=2$ 合法, 但定义域检查不能省略。

05.2

Calculus: Attach Meaning to Every Symbol

微积分：每个符号都要有题目含义

$f'(x) = 0$ gives only a stationary-point candidate, not automatically a maximum. Use derivative sign, the second derivative or domain endpoints. In optimisation, state the allowed x -range before comparing candidates.

$f'(x)=0$ 只给出驻点候选, 不自动保证最大值。需要结合导数变号、二阶导数或定义域端点判断。优化题尤其要先写允许的 x 范围, 再比较所有候选。

A definite integral gives signed area; a graph below the x -axis contributes negatively. For total geometric area, split at zeros and take positive areas of each part.

定积分给出带符号面积; 图像在 x 轴下方时贡献为负。题目若问几何总面积, 必须在过零点处分段并取各部分的正面积。

Stationary point 驻点	$f'(x) = 0$ gives a candidate location; its type still needs classification. $f'(x)=0$ 给出候选位置；类型仍需判断。 $f'(x) = 0$
Increasing 递增	On a differentiable interval, $f'(x) > 0$ means the function is increasing. 在可导区间内，$f'(x) > 0$ 表示函数递增。
Signed area 带符号面积	A definite integral is positive above the x-axis and negative below it. 定积分在 x 轴上方为正、下方为负。
Endpoints 端点	For extrema on a closed interval, compare interior stationary points with both endpoints. 闭区间上的最大最小值要同时比较内部驻点和两个端点。

ORIGINAL WORKED EXAMPLE Original Worked Example: State the Valid Range Before Optimising 原创例题：优化前先写有效范围	
PROBLEM	A rectangle symmetric about the y-axis lies under $y=12-x^2$ and above the x-axis. Its upper-right corner is $(x,12-x^2)$, where $x>0$. Find its maximum area. 一个关于 y 轴对称的矩形位于抛物线 $y = 12 - x^2$ 与 x 轴之间，右上角为 $(x, 12-x^2)$，其中 $x > 0$。求矩形的最大面积。
State dimensions and range 1. 写尺寸和范围	The rectangle has width $2x$ and height $12 - x^2$, requiring $0 < x < \sqrt{12}$. 矩形宽为 $2x$，高为 $12-x^2$；需要 $0 < x < \sqrt{12}$。
Write the area function 2. 写面积函数	$A(x) = 2x(12 - x^2) = 24x - 2x^3$. $A(x)=2x(12-x^2)=24x-2x^3$。 $A(x) = 24x - 2x^3$
Find the stationary point 3. 找驻点	$A'(x) = 24 - 6x^2$. Setting this to zero gives $x = 2$ within the valid range. $A'(x)=24-6x^2$。令其为 0，在有效范围内得到 $x=2$。 $A'(x) = 24 - 6x^2 = 0$
Classify and evaluate 4. 判断并求值	The derivative changes from positive to negative, so $x = 2$ gives a maximum; $A(2) = 4 \times 8 = 32$. 导数从正变负，所以 $x=2$ 给出最大值；$A(2)=4 \times 8=32$。
CONCLUSION	The maximum area is 32 square units. 最大面积为 32 平方单位。
COMMON TRAP	Writing $f'(x) = 0$ does not complete optimisation; the candidate must lie in the valid range and actually be a maximum. 只写 $f'(x)=0$ 还没有完成优化；必须确认候选在有效范围内并确实是最大值。

Why is $f(2)$ alone insufficient to identify the maximum on $[0,4]$?

若函数在 $[0,4]$ 上只有一个驻点 $x=2$ ，为什么不能只计算 $f(2)$ 就宣布最大值？

ANSWER / An absolute maximum can occur at an endpoint, so compare $f(0)$, $f(2)$ and $f(4)$.

答案 / 闭区间绝对最大值也可能出现在端点；必须比较 $f(0)$ 、 $f(2)$ 和 $f(4)$ ，并确认函数在区间内有定义。

CHAPTER 06 / TMUA FOUNDATIONS

Mathematical Reasoning and Option Discrimination

数学推理与选项辨析

Paper 2 often asks whether an argument is complete, a conclusion is necessary, or an option can be eliminated quickly, rather than requiring long calculation.

Paper 2 经常不要求长计算，而要求你判断一个论证是否完整、一个结论是否必然，或哪个选项能被条件迅速排除。

LEARNING OUTCOMES

- Distinguish experimental observation, general conjecture and rigorous proof

能区分实验观察、一般猜想与严格证明

- Use parity, remainders, bounds, symmetry and invariants to reduce possibilities

能使用奇偶、余数、边界、对称和不变量缩小可能性

- Choose the lowest-cost route among substitution, counterexample and full derivation across five options

能在五个选项中选择代入、反例或完整推导的最低成本路径

06.1

Must, May and Cannot

必然、可能与不可能

For must be true, one supporting example proves nothing; derive the conclusion from the conditions or eliminate all alternatives. For could be true, one construction satisfying every condition is enough.

题目问 must be true 时，一个支持例子没有证明力；你需要从条件推出结论，或证明所有其他选项不可能。题目问 could be true 时，一个满足全部条件的构造就足够。

For integers, check parity and modular structure; for geometry, bounds and symmetry; for functions, domain, monotonicity and extremes. An invariant can replace a long calculation.

整数问题优先看奇偶和模运算，几何问题优先看边界与对称，函数问题优先看定义域、单调和极端值。先找不变量，可以避免把题目变成长计算。

Must be true 必然成立	All valid cases must be covered; individual numerical examples are insufficient. 需要覆盖全部合法情况；个别数值例子不足。
Could be true 可能成立	One construction satisfying every premise is sufficient. 构造一个满足全部前提的例子即可。

Counterexample 反例	To refute a conclusion, the example must first satisfy every premise. 要推翻结论，例子必须先满足题目所有前提。
Invariant 不变量	Look for parity, remainder, sum or colour class preserved by the operation. 寻找操作前后保持不变的奇偶、余数、总和或颜色类别。

ORIGINAL WORKED EXAMPLE

Original Worked Example: Divisibility from Consecutive Integers

原创例题：用连续整数结构证明整除

PROBLEM	Prove that $n(n-1)(n+1)$ is divisible by 6 for every integer n . 证明：对任意整数 n ， $n(n-1)(n+1)$ 都能被 6 整除。
Recognise the structure 1. 识别结构	$n - 1$, n and $n + 1$ are three consecutive integers. $n-1$ 、 n 、 $n+1$ 是三个连续整数。
Find a factor of 2 2. 找因子 2	At least one of any two consecutive integers is even, so the product contains a factor of 2. 任意两个连续整数中至少一个是偶数，所以三者乘积含因子 2。
Find a factor of 3 3. 找因子 3	Exactly one of any three consecutive integers is divisible by 3, so the product contains a factor of 3. 任意三个连续整数中恰有一个能被 3 整除，所以乘积含因子 3。
Combine the factors 4. 合并	Because 2 and 3 are coprime, the product contains a factor of 6. 2 与 3 互质，因此乘积含因子 6。
CONCLUSION	$n(n - 1)(n + 1)$ is divisible by 6 for every integer n . $n(n-1)(n+1)$ 对所有整数 n 都能被 6 整除。
COMMON TRAP	Testing $n = 1, 2$ and 3 supports the conjecture but does not prove it for every integer. 代入 $n=1、2、3$ 只能支持猜想，不能证明对所有整数都成立。

ACTIVE RECALL / 合上笔记回答
What is the shortest valid way to show an integer expression is not always even? 要证明某个整数表达式不一定是偶数，最短的合法方法是什么？
ANSWER / Give one counterexample satisfying every premise for which the expression is odd. 答案 / 找一个满足所有前提、但使表达式为奇数的反例。

The five options are not decorative. If they differ in sign, interval or magnitude, use those differences to choose the cheapest test. Derive fully only when cheaper tests leave several options.

五个选项不是答案列表之外的装饰。若选项分布在不同符号、区间或数量级，先用这些差异决定最便宜的检验；只有当多个选项仍无法区分时才做完整推导。

A special value must be legal and discriminating. Zero, 1, -1, endpoints and symmetry points are useful, but if several options coincide there, choose another value or return to a general argument.

代入特殊值时必须选择合法且有区分度的值。0、1、-1、端点和对称点常有用，但如果它们让多个选项偶然相同，就需要换值或回到一般推导。

Sign 符号	Determine positive, negative or zero first; this often eliminates most options immediately. 先判断正、负或零，常能立即排除多数选项。
Magnitude 数量级	Use bounds or safe estimation to identify the interval containing the result. 用上下界或安全估算判断结果落在哪个区间。
Special value 特殊值	Substitute only within the allowed domain and confirm that the value separates the options. 只可在题目允许的定義域內代入，并确认该值能区分选项。
Reverse test 反向检验	Substitute options into necessary conditions when candidates are limited and forward solution is long. 从选项代回必要条件，适合候选值有限而正向求解较长的题。

ACTIVE RECALL / 合上笔记回答

If only one option is negative and the expression is certainly below zero, is full simplification needed?

若五个选项中只有一个为负数，而题目表达式显然小于 0，你还需要完整化简吗？

ANSWER / No. A rigorous sign argument is sufficient to select the only negative option.

答案 / 不需要。只要负号判断严格来自题目条件，就可以直接选择唯一负数选项。

Past-paper Training and Evidence-based Review

真题训练与证据化复盘

The value of past papers lies not in the number completed, but in whether each attempt leaves comparable data and a concrete next training action.

真题的价值不在于做过多少套，而在于每次尝试是否留下可比较的数据，并转化成下一次具体训练动作。

LEARNING OUTCOMES

- Decide when to use topic practice, a half-paper diagnostic or a complete timed paper

能决定何时做分主题训练、半套诊断或完整限时卷

- Choose the correct repair from the first error and timing record

能依据第一处错误和时间记录选择正确补救方式

- Use a transfer item after 24 hours and seven days to test genuine retention

能在 24 小时和 7 天后用迁移题检验是否真正掌握

07.1

Protect the Papers and Establish a Baseline

先保护真题，再建立基线

If the curriculum scope is incomplete, consuming a whole past paper mixes content gaps with test technique. Complete curriculum mapping and topic foundations before using one unseen paper for a timed baseline.

如果课程范围尚未覆盖，直接消耗整套真题只会把知识缺口和考试技巧混在一起。先完成课程映射与分主题基础练习，再用一套未见卷建立限时基线。

A repeat score on the same paper is not a new ability estimate because memory inflates it. Repetition checks repair; a new benchmark requires unseen or deliberately equated material.

同一套试卷的重做分数不能当作新的真实能力分数，因为记忆会抬高结果。重做用于检查修复；新的 benchmark 必须来自未见题或经过等值设计的模拟卷。

Unseen baseline 未见基线	The first complete timed attempt records starting accuracy, completion and pacing. 第一次完整限时作答，用于记录起始正确率、完成度和节奏。
Repair attempt 修复重做	Repeat an old question with explanations hidden to verify repair, not to create a new benchmark. 遮住解析重做旧题，用于验证错误是否被修复，不作为新 benchmark。
Transfer check 迁移检查	Use a new surface structure with the same knowledge to test whether only the original answer was memorised. 使用同知识、不同表面结构的新题，判断是否只记住原答案。
Paper preservation 保护真题	Assign each paper a purpose—baseline, progress check or final simulation—before opening it. 给每套卷指定用途：基线、阶段检查或最终模拟，避免无目的打开。

Why does improving the same paper from 22/40 to 36/40 one week later not prove a 36/40 current ability?

为什么一周后把同一套卷从 22/40 提高到 36/40，不能直接宣称能力达到 36/40？

ANSWER / The second score includes memory and solution exposure; use unseen or equated material for a fresh estimate.

答案 / 第二次结果混入了题目记忆和解析暴露。它能证明部分修复，但新的能力估计需要未见题或等值模拟卷。

07.2

Turn Error Labels into the Next Training Action

从错误标签到下一次训练动作

Review the first point where the answer diverged, not only the final line. If the conditional direction was misread, repair qualifiers and arrows even if later algebra also failed. If the route was correct but a sign was copied wrongly, train a checkpoint rather than relearning the chapter.

复盘先找第一处导致答案偏离的节点，而不是只看最后一行。读题方向错了，即使后面代数也错，首要修复仍是限定词和条件箭头；反之，路径完全正确却抄错符号，应训练检查点而不是重学整章。

Separate time into reading, active thinking and execution. Long time does not automatically mean weak knowledge; the route may be missing or the options underused. Timing becomes meaningful only beside the scratch work and first move.

时间数据要分成阅读、主动思考和执行。长时间并不自动代表知识差：可能是启动路径缺失，也可能是选项利用不足。只有结合草稿和第一步记录，时间才有解释价值。

Knowledge action 知识缺口	Learn the minimum concept unit, then complete two or three direct applications. 学习最小概念单元，随后做 2 至 3 道直接应用题。
Route action 路径缺口	Write the trigger and first move, then practise the same method under different surface forms. 写出触发信号与第一步，再做不同表面的同方法题。
Execution action 执行错误	Add sign, range, transcription or estimation checks at high-risk steps. 在高风险步骤设置符号、范围、抄写或估算检查点。
Reading action 读题错误	Circle qualifiers such as necessary, sufficient, must, could, integer and distinct, then restate the task. 圈出 necessary、sufficient、must、could、integer、distinct 等限定词并复述题意。

ORIGINAL WORKED EXAMPLE

Original Worked Example: Identify the Real Training Gap

原创例题：识别真正的训练缺口

PROBLEM

A student knows the discriminant but cannot start a parameter question after 4 minutes 40 seconds. One trigger sentence enables a complete solution, yet a new parameter question causes the same failure seven days later. How should this be classified and trained?

一名学生会使用判别式，但在参数题上主动思考 4 分 40 秒仍没有第一步。看完一句“把不同实根翻译成 $\Delta > 0$ ”后可以独立完成；七天后遇到新参数题又无法启动。应怎样归因并安排下一步？

Rule out a knowledge gap 1. 排除知识缺口	The student can calculate the discriminant and complete the problem independently after a trigger, so the formula itself is not the main problem. 学生会计算判别式，也能在提示后独立完成，所以主要问题不是公式未知。
Locate the first failure 2. 找第一处失败	Failure occurs before translating the wording into a mathematical condition, so it is a route-start gap. 失败发生在把题目语言转换为数学条件之前，属于启动路径缺口。
Interpret the seven-day result 3. 解释七天后结果	Success on the old item but failure to start a new one shows that the surface solution was remembered without a transferable trigger. 旧题会做但新题不能启动，说明记住了解法表面，没有形成可迁移触发信号。
Assign the next action 4. 安排下一步	Create a roots-count => discriminant-condition trigger card and interleave three differently worded parameter items, recording only the translation and first move. 制作一张“根的数量 => 判别式条件”触发卡，并穿插完成3道不同表面的参数题；每题只记录题意翻译和第一步。
CONCLUSION	Classify this as a route gap. Train the trigger and cross-question transfer rather than relearning the complete quadratic chapter. 归类为 route gap；下一步训练触发信号和跨题型迁移，而不是重学完整二次方程章节。
COMMON TRAP	Treating a long time as proof of missing knowledge assigns too much content teaching without repairing the true starting problem. 看到用时很长就直接判断“知识不会”，会安排过量课程，却没有修复真正的启动问题。

ACTIVE RECALL / 合上笔记回答

A student reads distinct roots as real roots and uses delta greater than or equal to zero. What are the first error label and next action?

一道题因把 distinct roots 误读成 real roots 而使用 $\Delta \geq 0$ 。第一错误标签和下一步动作是什么？

ANSWER / A reading error; map distinct to the strict condition delta greater than zero and check the inequality before submission.

答案 / 读题错误。建立限定词检查：distinct 对应严格条件 $\Delta > 0$ ，并在提交前复核不等号。

15-minute Active Recall Check

15 分钟主动回忆检查

Close the explanations and complete these independently first. Every question is an original Mantou item and does not correspond to an official past-paper question.

先合上解析独立完成。所有题目均为满托原创，不对应任何官方真题。

01	Which statement is equivalent to ‘A is necessary for B’? 设 A、B 为两个陈述。“A 是 B 的必要条件”等价于哪一项？
	- A. $A \Rightarrow B$ - B. $B \Rightarrow A$ - C. $\text{not } A \Rightarrow B$ - D. A if and only if B
ANSWER	B. B requires A, so $B \Rightarrow A$. B 成立必须有 A，所以方向是 $B \Rightarrow A$。

02	Which value is a counterexample to ‘for every real x, $x^2 \geq x$’? “对所有实数 x，$x^2 \geq x$”这一命题为假。哪一个数本身就构成反例？
	- A. -1 - B. 0 - C. $1/2$ - D. 2
ANSWER	C. At $x=1/2$, $x^2=1/4 < 1/2$. $x = 1/2$ 时，$x^2 = 1/4 < 1/2$。一个合法反例即可推翻全称命题。

03	What is the real solution of $\sqrt{x+3}=x-1$? 实数方程 $\sqrt{x+3} = x-1$ 的解是？
	- A. $(3-\sqrt{17})/2$ only - B. $(3+\sqrt{17})/2$ only - C. both values - D. no real solution
ANSWER	B. The original equation requires $x \geq 1$, so the negative candidate is extraneous. 原式要求 $x \geq 1$；负的候选根不满足限制。

04	For two distinct real roots, which condition on k is required? 方程 $x^2 - 2(k+1)x + k^2 + 1 = 0$ 有两个不同实根。k 必须满足？
	- A. $k < 0$ - B. $k \leq 0$ - C. $k > 0$ - D. all real k
ANSWER	C. The discriminant is $8k$, which must be positive. 判别式为 $8k$；两个不同实根要求 $8k > 0$。
05	Which statement is logically equivalent to $P \Rightarrow Q$? 下列哪一项与命题 $P \Rightarrow Q$ 逻辑等价？
	- A. $Q \Rightarrow P$ - B. $\text{not } P \Rightarrow \text{not } Q$ - C. $\text{not } Q \Rightarrow \text{not } P$ - D. $P \Rightarrow \text{not } Q$
ANSWER	C. A statement is equivalent to its contrapositive. 原命题与逆否命题 $\text{not } Q \Rightarrow \text{not } P$ 等价。
06	How is $y=f(x-4)$ obtained from $y=f(x)$? 图像 $y = f(x-4)$ 由 $y = f(x)$ 怎样得到？
	- A. ■■■ 4 - B. ■■ 4 - C. ■■ 4 - D. ■■■■■■ $1/4$
ANSWER	B. Replacing x by $x-4$ shifts the graph four units right. 括号内 $x-4$ 对应向右平移 4。可用原图上的特征点验证。
07	A positive geometric sequence has first term 3 and third term 12. What is the sum of its first four terms? 一个各项均为正数的等比数列首项为 3、第三项为 12。前四项之和是多少？
	- A. -15 - B. 21 - C. 45 - D. 48

ANSWER	C. Positivity selects $r=2$, giving $3+6+12+24=45$. $3r^2=12$ 给出 $r=\pm 2$; 各项为正排除 -2, 所以前四项为 3、6、12、24, 总和 45。
08	The line $y=mx+5$ is tangent to $x^2+y^2=5$. Which values can m take? 直线 $y=mx+5$ 与圆 $x^2+y^2=5$ 相切。m 的取值是?
	- A. 2 only - B. -2 only - C. 2 or -2 - D. 5 or -5
ANSWER	C. Equating the centre-to-line distance to the radius gives $m^2=4$. 原点到直线 $mx-y+5=0$ 的距离等于 $\sqrt{5}$, 平方后得到 $m^2=4$。
09	What is the solution of $2^{(x+1)}=8$ times 4^x? 方程 $2^{(x+1)}=8 \times 4^x$ 的解是?
	- A. -3 - B. -2 - C. 2 - D. 3
ANSWER	B. Rewrite the right side as $2^{(2x+3)}$, then equate exponents. 右侧为 $2^{(2x+3)}$, 所以 $x+1=2x+3$, 得到 $x=-2$。
10	A rectangle has width $2x$ and height $12-x^2$. For valid positive x, what is its maximum area? 矩形宽为 $2x$、高为 $12-x^2$。若 $x>0$ 且矩形有效, 其最大面积是多少?
	- A. 24 - B. 30 - C. 32 - D. 36
ANSWER	C. The valid stationary point $x=2$ gives area 32 and the derivative changes from positive to negative. $A=24x-2x^3$, $A'=24-6x^2$; 有效驻点 $x=2$ 给出最大面积 32。
11	Why is $n(n-1)(n+1)$ divisible by 6 for every integer n? 为什么 $n(n-1)(n+1)$ 对每个整数 n 都能被 6 整除?

	- A. ■■■■■■■■ - B. ■■■■■■■■■■■■■■■■■■ 3 ■■■■ - C. ■■ $n=1 \times 2 \times 3$ ■■■■ - D. n^2-1 ■■ 6 ■■■■
ANSWER	B. Among three consecutive integers one supplies a factor 2 and one a factor 3. 三个连续整数的乘积一定含因子 2 和因子 3；2 与 3 互质，所以乘积能被 6 整除。

12	A student's score rises from 22/40 to 36/40 on the same paper. What is the best next step for a fresh ability benchmark? 学生重做同一套卷从 22/40 提升到 36/40。若要获得新的能力 benchmark，下一步最合适的是？
	- A. ■■■■ 36/40 ■■■■■■ - B. ■■■■■■■■■■ - C. ■■■■■■■■■■■■■■■■■■ - D. ■■■■■■■■
ANSWER	C. A repeat score includes memory; use unseen or equated material for a fresh estimate. 重做分数混入题目记忆，适合验证修复；新的能力估计应来自未见题或等值模拟卷。

Turn Every Attempt into Evidence

把每道题变成下一次会做

01	Before 做题前	Label the topic and expected first move without checking an answer. 标记题目主题与预期第一步，不查答案。
02	During 做题中	Record active-thinking time, answer changes and whether elimination was used. 记录主动思考时间、改答和是否使用排除法。
03	After 提交后	Locate the first error rather than treating the final wrong step as the only cause. 定位第一处错误，不把最后一步错当作唯一原因。
04	24 hours later 24 小时后	Hide the explanation and repeat; review is complete only when the trigger can be explained. 遮住解析重做；能解释触发信号才算完成复盘。
05	7 days later 7 天后	Use one original transfer item with the same method rather than memorising the old answer. 用一道同方法原创题检查迁移，而不是背原题答案。

SOURCES

Version Boundary and Official Anchors

版本边界与官方依据

This resource independently reorganises and explains the public exam framework. All examples, checkpoint questions and explanations are original Mantou work and are not official questions; official materials are used only to verify scope and facts.

本资料依据公开考试框架重新组织并原创撰写。例题、自测题与解析均非官方真题；官方资料只用于核对考试范围与事实。

After the Foundation Edition, use the Six-Week Precision Training Plan to structure the work, then collect accuracy, timing and error evidence in past papers. Worked Explanations identify the reasoning worth repairing first.

完成基础版后，使用六周精确训练计划安排节奏，并在历年真题中积累正确率、用时和错误类型证据；逐题深度解析用于定位最值得修复的思路。

TMUA Content Specification — October 2026 and January 2027 Exam structure, Paper 1 and Paper 2 scope, and knowledge boundaries. 考试结构、Paper 1 / Paper 2 范围与知识点边界 Click the title to open the official source. 点击标题打开官方原文	Notes on Logic and Proof Paper 2 logic, quantifiers, proof methods and reasoning-error boundaries. Paper 2 逻辑、量词、证明方式与辨错范围 Click the title to open the official source. 点击标题打开官方原文
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Notes on Mathematics for TMUA Paper 1 and ESAT Mathematics 2

Mathematical terminology, scope boundaries and checks of common concepts.

数学术语、范围边界与常见概念核对

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UAT-UK Candidate Handbook — 2027 Entry

Live-test interface, separately timed papers and test-centre operation facts.

正式考试界面、分卷计时与考场操作事实

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NEXT TRAINING STEP

下一步训练

Complete active recall / practise by topic online / establish a baseline on an unseen paper

完成主动回忆检查 / 进入在线分主题练习 / 使用未见卷建立基线